

CMRDF: Morphology-Conditioned Reachability Fields for Task-Space Robot Policy Transfer

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Abstract: Generalist robot policies often represent actions as end-effector target poses in task space, which facilitates policy transfer across robots performing the same task. However, a task-space pose that is feasible for one morphology may be infeasible for another due to joint limits, workspace differences, or self-collision constraints. Existing pose reachability methods often learn monolithic fields over the full $SE(3)$ pose space, which spreads expensive inverse-kinematics supervision over a high-dimensional domain and leaves near-boundary regions poorly sampled in cross-morphology settings. We propose Cross-Morphology Reachability Distance Fields (CMRDF), a morphology-conditioned task-space reachability representation that decomposes full-pose feasibility into global position reachability and position-conditioned orientation reachability. The two outputs are signed-distance function (SDF) proxies with physical units. CMRDF uses dense forward-kinematics coverage and distance-to-boundary targets for the Position-SDF, and inverse-kinematics boundary mining at representative position anchors for the Orientation-SDF. The differentiable fields support reachability queries and gradient-based local pose repair. Under matched full-pose inverse-kinematics budgets, CMRDF improves $SE(3)$ accuracy over a capacity-matched monolithic baseline by 2.5 percentage points at both $1.00\times$ and $0.51\times$. It also supports low-distortion pose repair and zero-shot transfer under joint-limit and same-family geometry shifts. In target-side policy fine-tuning, converted demonstrations reduce the required native data from 60–80 to 20 demonstrations at 90% of a target-only reference policy’s success rate.

Keywords: Morphology-conditioned reachability, $SE(3)$ pose repair, Task and motion planning

1 Introduction

Many task-space robot policies and generalist action decoders output end-effector poses, pose increments, or short-horizon action segments that implicitly determine end-effector motion. Separating task planning from the specific execution morphology in this way facilitates the generalization of the same policy across morphologies. However, task-space planning itself does not encode the kinematic information of the execution morphology: a target pose that is reachable on the training morphology may violate joint limits or self-collision constraints on another morphology. This is one reason why generalist robot policies, including vision-language-action (VLA) models, often still require additional morphology adaptation when transferred to new robot morphologies [1, 2, 3, 4, 5]. To make cross-morphology transfer effective at the execution layer, we need a morphology-aware task-space reachability prior that can quickly determine whether a target is executable and, when necessary, locally repair the pose.

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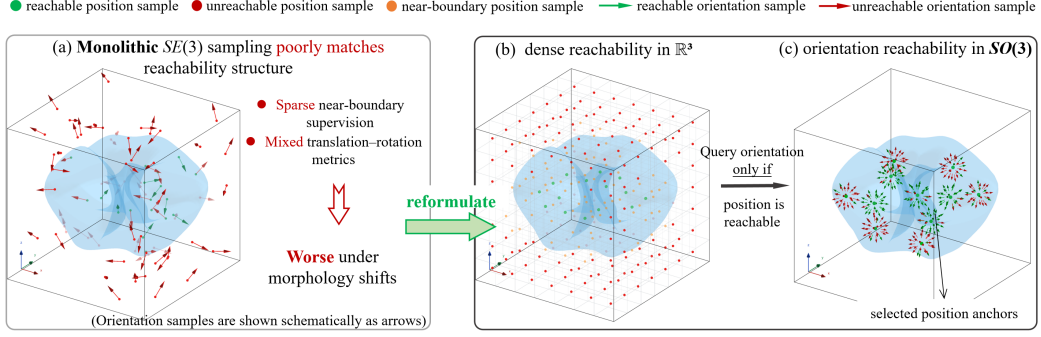


Figure 1: CMRDF reformulates morphology-conditioned pose reachability as a position-first hierarchical decomposition: a monolithic $SE(3)$ field is factorized into a dense Position-SDF over $\mathbb{R}^3$ and a position-conditioned Orientation-SDF over $SO(3)$ at selected reachable position anchors.

Existing discrete or continuous reachability representations for full-pose queries usually merge position and orientation into a monolithic $SE(3)$ function [6, 7, 8, 9]. This representation not only increases the search dimension, but also mixes the different metric structures of Euclidean position and $SO(3)$ orientation, making it difficult for limited supervision to concentrate near the reachability boundary that determines decision and repair quality. In the cross-morphology setting, the boundary also changes with the robot morphology, further reducing the supervision efficiency of monolithic representations.

Our starting point is a position-first decomposition of pose reachability: a pose is reachable if and only if its position lies inside the reachable position domain and its orientation belongs to the position-conditioned reachable orientation set at that position. Based on this decomposition, we use two conditionally coupled boundary fields rather than a single scalar field over $SE(3)$: a Position-SDF proxy that describes the reachable boundary at the position layer, and an Orientation-SDF proxy that describes the orientation reachability boundary at a given position. Once the two are separated, the supervision budget, output semantics, and gradients used by pose repair can all be aligned with the geometry of the problem.

Based on this structure, we propose Cross-Morphology Reachability Distance Fields (CMRDF). CMRDF conditions on kinematic topology, link geometry, and self-collision patterns through a shared morphology embedding, and reallocates the expensive full-pose IK budget from global $SE(3)$ sampling to orientation boundary mining around a small number of informative position anchors. Because its outputs are signed-distance proxies with physical units, CMRDF can serve as a fast pose reachability prior and provide local gradient signals for low-distortion sequential pose repair.

The contributions of this paper are as follows:

1. We propose a morphology-conditioned distance-field representation for cross-morphology $SE(3)$ pose reachability, reformulating monolithic 6D reachability learning as the joint modeling of a Position-SDF and a position-conditioned Orientation-SDF.
2. We propose a position-dense, orientation-focused supervision strategy aligned with this representation, combining dense FK/EDT position supervision with IK boundary mining at a small number of anchors to improve near-boundary modeling efficiency without exhaustive global $SE(3)$ enumeration.
3. At matched full-pose IK budgets, CMRDF improves prediction over a capacity-matched monolithic baseline. Separate experiments demonstrate low-distortion local pose repair, zero-shot transfer under joint-limit and same-family geometry shifts, and downstream data utility for real-robot policy fine-tuning.

2 Related Work

Discrete and continuous task-space reachability representations. Task-space reachability representations can be broadly divided into discrete maps and continuous predictors. Discrete methods precompute workspace or task-space reachability offline, enabling efficient queries. However, they are limited by resolution, storage, and construction cost, and they usually lack continuous differentiable boundary signals [6, 7, 10, 11, 12, 13]. Continuous methods learn reachability functions to reduce lookup and repeated IK [8, 9]; representative methods usually take a pose query as a joint task-space input or store orientation-related reachability through orientation bins. In contrast, CMRDF explicitly decomposes pose reachability into a Position-SDF proxy and a position-conditioned Orientation-SDF proxy, and concentrates IK supervision near orientation reachability boundaries. It is related in form to neural implicit fields, but its modeling target is pose reachability rather than geometric surfaces [14, 15, 16, 17].

Cross-morphology continuous workspace and capability representations. Recent work has explored cross-morphology workspace or capability fields and has shown generalization to held-out robots [18]. However, such methods mainly operate at the workspace position level and do not handle position-conditioned orientation reachability in full $SE(3)$ poses. This is the gap addressed by this paper: CMRDF explicitly decomposes pose reachability into a Position-SDF and a position-conditioned Orientation-SDF, thereby modeling the orientation boundary in full-pose queries.

Cross-morphology shared representations and generalist robot policies. Existing modular, graph-based, and structure-aware policies, as well as cross-morphology imitation, demonstrate the feasibility of sharing structured representations across morphologies [19, 20, 21, 22, 23, 24]. RT-1, RT-2, Open X-Embodiment, Octo, and OpenVLA further advance generalist robot policies and VLA models [1, 2, 3, 4, 5]. These works usually focus on shared policy learning and do not directly provide an explicit, differentiable task-space reachability prior. CMRDF is complementary to them: it does not replace the upper-level policy, but provides a morphology-conditioned geometric interface between high-level pose proposals and embodiment-specific execution.

3 Method

This section presents Cross-Morphology Reachability Distance Fields (CMRDF). CMRDF represents cross-morphology pose reachability as two conditionally coupled signed-distance proxies: a Position-SDF that describes the reachable position domain, and an Orientation-SDF that describes the position-conditioned reachable orientation set at a given position.

3.1 Problem Structure and Two-Field Representation

Given a serial manipulator morphology $\mathcal{M}$, let $\mathcal{Q}(\mathcal{M})$ denote the joint space satisfying joint limits, and let f_{FK} denote forward kinematics. The **intrinsic pose reachability** considered in this paper is determined by joint limits and self-collision constraints, and does not include environmental obstacles, dynamics, or contact constraints. The corresponding reachable pose set is defined as

$$\mathcal{R}(\mathcal{M}) = \{(p, R) \in SE(3) \mid \exists \mathbf{q} \in \mathcal{Q}(\mathcal{M}), f_{FK}(\mathbf{q}) = (p, R), \mathbf{q} \text{ satisfies self-collision constraints}\}. \quad (1)$$

This set induces a reachable position domain $\mathcal{P}(\mathcal{M}) \subset \mathbb{R}^3$ and a position-conditioned reachable orientation set $\mathcal{R}_p(\mathcal{M}) \subset SO(3)$. For any query pose $x = (p, R)$, we have

$$x \in \mathcal{R}(\mathcal{M}) \iff p \in \mathcal{P}(\mathcal{M}) \wedge R \in \mathcal{R}_p(\mathcal{M}). \quad (2)$$

Either conditioning order is mathematically valid. We choose position-first to combine dense FK/EDT supervision in 3D with IK orientation-boundary mining at selected positions. CMRDF learns two signed-distance proxies from boundaries estimated by the Stage-I and Stage-II numerical supervision

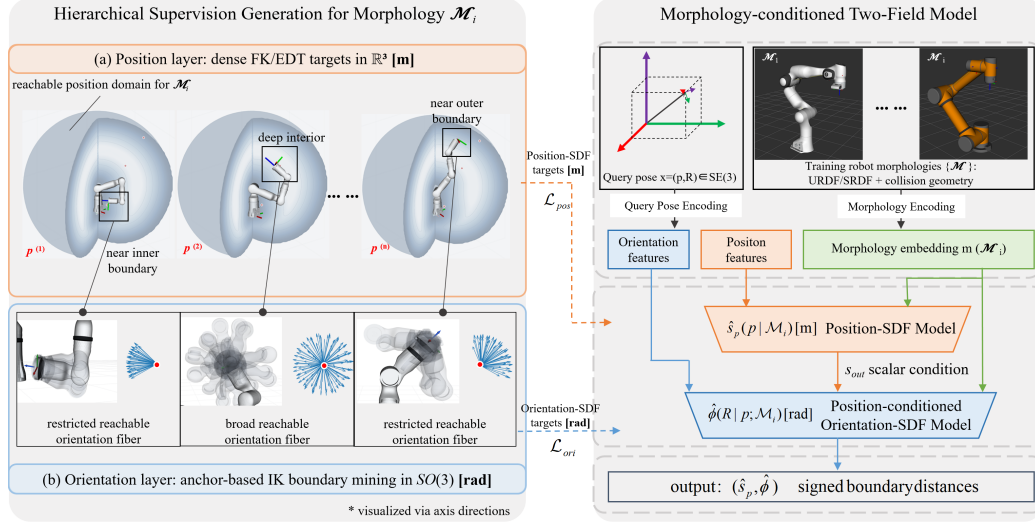


Figure 2: CMRDF supervision and morphology-conditioned two-field architecture. Dense FK/EDT supervision is used for the Position-SDF, and anchor-based IK mining is used for the Orientation-SDF.

procedures. Let the estimated position boundary be $\widetilde{\partial\mathcal{P}}(\mathcal{M})$ and the estimated orientation boundary be $\widetilde{\partial\mathcal{R}}_p(\mathcal{M})$. The corresponding proxy targets are

$$\tilde{s}_p(p | \mathcal{M}) = \tilde{y}_p(p) \min_{p_b \in \widetilde{\partial\mathcal{P}}(\mathcal{M})} \|p - p_b\|_2, \quad \tilde{\phi}(R | p; \mathcal{M}) = \tilde{y}_R(R, p) \min_{R_b \in \widetilde{\partial\mathcal{R}}_p(\mathcal{M})} d_{SO(3)}(R, R_b). \quad (3)$$

Here, $\tilde{y}_p, \tilde{y}_R \in \{+1, -1\}$ are signs assigned by the conservative FK/EDT position supervision in Stage-I and the multi-start IK orientation mining in Stage-II, respectively. The network predictions $\hat{s}_p$ and $\hat{\phi}$ approximate these proxy targets; a positive value indicates reachability at the corresponding layer under the numerical oracle, while a negative value indicates infeasibility. The two predictions have physical units of [m] and [rad], respectively. Since the orientation field has geometric meaning only when the position is reachable, CMRDF adopts position-first logic in both querying and repair.

3.2 Position-Dense, Orientation-Focused Hierarchical Supervision

The hierarchical reformulation directly determines how the supervision budget is allocated. Stage-I uses low-cost FK coverage to construct dense position supervision: for each morphology, self-collision-free FK hits are collected, reachable positions are voxelized in a morphology-dependent working domain, and a conservative boundary is extracted from inside–outside neighborhoods. Frontier position-DLS completion is then used to reduce local holes caused by finite sampling, and a 3D EDT is computed on the conservative boundary to obtain the Position-SDF proxy target. This stage covers the position space at low cost while reducing false positive reachability through the conservative boundary.

Stage-II concentrates expensive IK queries on a small number of informative position anchors. At each anchor, CMRDF performs multi-resolution orientation certification over $SO(3)$, labels local orientation cells with a multi-start IK oracle, extracts boundary representatives from mixed regions and inside/outside adjacency, and then generates boundary, shell, and global Orientation-SDF samples. For anchors verified as full-reachable, it applies only positive lower-bound supervision with $\tilde{\phi}_{full} = \pi$. In this way, orientation supervision is concentrated near the position-conditioned orientation boundary rather than uniformly spending full-pose IK budget over the entire $SE(3)$.

3.3 Morphology-Conditioned Two-Field Representation

CMRDF encodes kinematic topology, link geometry, joint limits, and self-collision priors into a shared morphology embedding $m_{\mathcal{M}}$ through a graph-based encoder, which conditions both fields.

The Position branch normalizes the query position by the morphology-dependent reference length $L_{\text{ref}}(\mathcal{M})$, predicts the normalized Position-SDF, and also outputs a local position context for the orientation layer. The physical distance is recovered by denormalization:

$$(\hat{s}_p, \hat{z}_{\text{pos}}) = f_{\theta_p}(E_p(\tilde{p}), m_{\mathcal{M}}), \quad \hat{s}_p(p | \mathcal{M}) = L_{\text{ref}}(\mathcal{M})\hat{s}_p(\tilde{p} | \mathcal{M}). \quad (4)$$

Here, $\hat{z}_{\text{pos}}$ is the anchor-level predicted local position context produced by the Position branch, with training supervision from Stage-I position statistics.

The Orientation branch uses a morphology-conditioned local-context FiLM-SIREN decoder [25, 16]. Let $\hat{z}_c$ and $\hat{z}_l$ denote the center context and local patch context sampled from $\hat{z}_{\text{pos}}$ and its local neighborhood, respectively. The orientation-layer interface is

$$\hat{\phi}_{\text{raw}} = f_{\theta_R}(E_R(R), E_p(\tilde{p}), \text{clip}(\hat{s}_p, -1, 1), \hat{z}_c, \hat{z}_l, m_{\mathcal{M}}), \quad \hat{\phi} = \pi \tanh\left(\frac{\hat{\phi}_{\text{raw}}}{\pi}\right). \quad (5)$$

Here, E_R is consistent with quaternion double cover; $\hat{\phi}_{\text{raw}}$ preserves unsaturated gradients, while $\hat{\phi}$ is a bounded radian output. The coverage head is used only as a training auxiliary.

3.4 Training and Inference Interface

Training has two stages: Stage-A learns the Position-SDF and local context, and Stage-B learns the position-conditioned Orientation-SDF. Both use distance regression, sign supervision, and Eikonal regularization, with additional regularizers for full-reachable anchors, far negatives, and cross-anchor continuity.

At inference time, CMRDF first queries the position layer, and then queries the orientation layer when the position is reachable:

$$\widehat{\text{Reach}}(p, R; \mathcal{M}) = \mathbf{1}[\hat{s}_p(p | \mathcal{M}) > 0] \cdot \mathbf{1}[\hat{\phi}(R | p; \mathcal{M}) > 0]. \quad (6)$$

If a query pose is infeasible, this paper uses sequential projection for local repair: it first updates the position along the physical gradient of the Position-SDF until the position enters the reachable inner band; it then fixes the position and updates the orientation along the gradient of $\hat{\phi}_{\text{raw}}$ in the local Lie-algebra coordinates of $\text{SO}(3)$. The orientation update uses the raw field to avoid gradient attenuation caused by $\tanh$ saturation. The task weights in Sec. 4.3 are used only for evaluation and do not enter the online repair update.

4 Experiments

Experiments evaluate reachability prediction under matched full-pose IK budgets, low-distortion local pose repair, zero-shot transfer under joint-limit and same-family geometry shifts, and downstream data utility for robot policy fine-tuning.

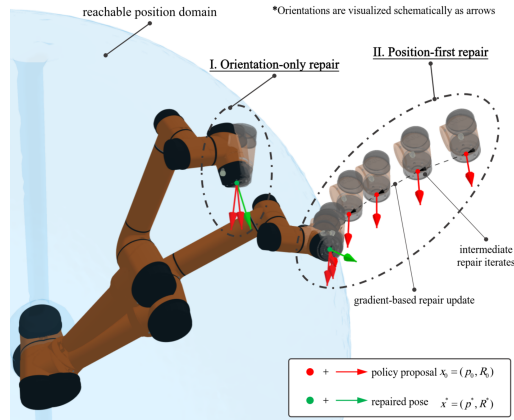


Figure 3: Typical local repair behaviors of CMRDF: orientation-only correction, and position-first correction followed by orientation refinement.

4.1 Setup and Evaluation Protocols

We construct a morphology suite consisting of real base morphologies and controlled same-family variants. The training set contains 5 real serial manipulator base morphologies. For each base, we further generate 3 joint-limit variants and 3 same-family geometry variants, yielding 35 training morphologies in total. The former changes the available joint ranges, while the latter preserves topology and joint order but changes link lengths, with synchronized updates to kinematic transforms, collision geometry, URDF/SRDF, and the MoveIt setup [26].

Secs. 4.2–4.4 evaluate pose prediction, near-boundary repair, and structured-shift transfer using a common decision oracle: MoveIt IK success, joint-limit satisfaction, and absence of self-collision. The full-pose IK supervision budget in Sec. 4.2 excludes FK/EDT position supervision, frontier position-DLS completion, and model training, and is not a measure of total offline compute. Sec. 4.5 separately evaluates demonstration conversion and policy fine-tuning through real-robot task success and native-data requirements.

4.2 Matched-Budget Reachability Prediction

We evaluate held-out pose queries on the five base morphologies included in training. Table 1(a) compares DRM-NN and DRM-RSVM trained on MonoData or HierData, and CMRDF at three orientation-anchor budgets. Anchor-Seen uses held-out orientations at supervised anchors; Anchor-Heldout uses positions excluded from orientation supervision. Table 1(b) uses DeepMono-SE(3), with the same Graph-Transformer morphology encoder architecture and pose encodings, a monolithic decoder from the same FiLM-SIREN family, and a total trainable parameter count matched to the complete CMRDF model. Within each full-pose IK budget, optimization budgets and fixed test sets are matched. Appendix B.1 provides baseline and split details.

Table 1: Pose reachability on five training base morphologies. (a) DRM and anchor-budget evaluation: macro averages. (b) Capacity-matched evaluation: mean $\pm$ standard deviation over three seeds, after macro-averaging over the five bases. NB denotes near-boundary accuracy; latency in (a) is reported in milliseconds.

(a) DRM comparison and anchor-budget sweep							
Method	Supervision	Budget		Accuracy (%) $\uparrow$			Latency $\downarrow$
		Anch.	IK	SE(3)	NB-Seen	NB-Heldout	ms
DRM-NN	MonoData	–	1.00 $\times$	74.86	62.88	62.57	0.114
DRM-NN	HierData	2048	1.00 $\times$	79.08	67.25	65.43	0.083
DRM-RSVM	MonoData	–	1.00 $\times$	74.59	63.71	63.17	1.470
DRM-RSVM	HierData	2048	1.00 $\times$	78.86	68.14	66.03	1.520
CMRDF	HierData	2048	1.00 $\times$	92.04	90.27	87.18	0.182
CMRDF	HierData	1640	0.79 $\times$	90.48	87.53	84.32	0.179
CMRDF	HierData	1024	0.51 $\times$	85.53	74.96	69.47	0.181
(b) Capacity-matched comparison							
Method	Full-pose IK budget		SE(3)-Acc. (%) $\uparrow$		NB-Heldout (%) $\uparrow$		
DeepMono-Mono	1.00 $\times$		86.3 $\pm$ 2.6		80.7 $\pm$ 2.3		
DeepMono-Hier	1.00 $\times$		89.5 $\pm$ 2.1		84.1 $\pm$ 2.1		
CMRDF	1.00 $\times$		92.0 $\pm$ 0.3		87.2 $\pm$ 0.5		
DeepMono-Hier	0.51 $\times$		82.9 $\pm$ 3.1		66.5 $\pm$ 2.9		
CMRDF	0.51 $\times$		85.4 $\pm$ 0.4		69.8 $\pm$ 0.8		

Hierarchical supervision improves the evaluated DRM learners on both near-boundary splits. In the anchor-budget sweep, CMRDF’s NB-Heldout accuracy increases from 69.47% at 0.51 $\times$ to 84.32% at 0.79 $\times$ and 87.18% at 1.00 $\times$; Figure 4(a) and Table 5 show the per-base behavior. The Seen–Heldout gap widens as the anchor budget falls, while reported query latency remains within

0.179–0.182 ms. In the capacity-matched comparison, HierData improves DeepMono by 3.2/3.4 percentage points in SE(3)/NB-Heldout accuracy at $1.00\times$. With HierData matched, CMRDF further gains 2.5/3.1 points at $1.00\times$ and 2.5/3.3 points at $0.51\times$ (Figure 4(b)). Reduced-budget CMRDF remains below full-budget DeepMono-Hier. These results support the complete two-field design with its field-specific supervision and geometric constraints, not hierarchical factorization alone.

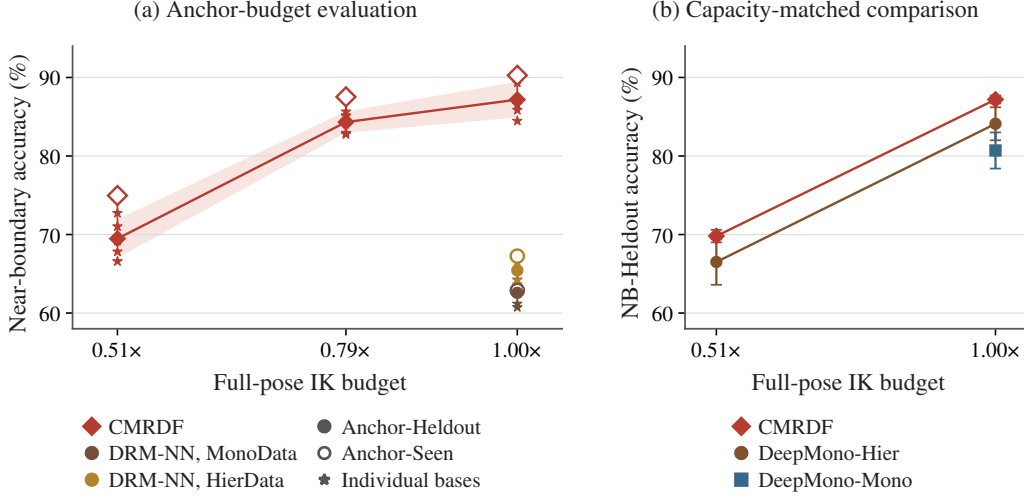


Figure 4: Near-boundary accuracy versus full-pose IK budget. (a) Anchor-budget evaluation: filled/open markers show Anchor-Heldout/Seen means, small stars individual bases, and shading one standard deviation across bases. (b) Capacity-matched evaluation: markers and error bars show the three-seed mean $\pm$ standard deviation of five-base macro averages. Lines connect measured budgets only.

4.3 Distance Semantics for Low-Distortion Pose Repair

This subsection further examines whether distance semantics truly improves local repair, rather than only improving decisions. The benchmark uses only near-boundary infeasible queries, and reports results under two task-weighted evaluation regimes: one that prioritizes preserving position and one that prioritizes preserving orientation. The main metric is the task-weighted repair cost J_{norm} , and Gap-to-Ref is computed against Ref-IK, a high-budget reference solution.

We compare five online methods: DRM-Score, Hierarchical-Cls, Search-IK, CMRDF w/o Ori-Eik, and CMRDF. Ref-IK is used only as a high-budget reference solution.

Table 2: Near-boundary pose repair benchmark. Macro averages over five base morphologies and two task-weight regimes; success is oracle-verified, Gap is relative to Ref-IK, and latency excludes final validation.

Method	Repair	Distortion $\downarrow$		Cost $\downarrow$		Time $\downarrow$
	Succ. (%) $\uparrow$	$\Delta p/L_{\text{ref}}$	ΔR (rad)	J_{norm}	Gap	ms
DRM-Score	85.4	0.051	0.245	0.0348	0.0238	1.52
Hierarchical-Cls	88.3	0.040	0.203	0.0289	0.0179	1.33
Search-IK	93.1	0.023	0.114	0.0176	0.0067	13.84
CMRDF w/o Ori-Eik	92.2	0.016	0.094	0.0144	0.0035	1.43
CMRDF	94.7	0.010	0.069	0.0118	0.0009	1.31
Ref-IK (high-budget)	97.1	0.008	0.064	0.0109	0.0000	34.57

Table 2 compares feasibility recovery and distortion across the evaluated repair methods. DRM-Score and Hierarchical-Cls have larger repair costs and Gap-to-Ref, while Search-IK improves success over these methods at higher latency. Among the evaluated online methods, CMRDF has the highest oracle-verified success rate and the lowest repair cost and Gap-to-Ref. The comparison with CMRDF w/o Ori-Eik further supports the role of orientation-gradient regularization in low-distortion repair.

4.4 Zero-Shot Transfer under Structured Morphology Shifts

We evaluate whether the shared morphology-conditioned representation can adapt to structured morphology shifts (Limit-shift and Family-shift) without resampling data or retraining the model for the target morphology, and we report a Real-ODD stress test as a stronger out-of-support diagnostic. We use SE3-Acc to measure overall decisions, and $\text{Rec}@U_{\text{nb}}$ to measure the end-to-end ability to first correctly reject and then repair back to reachability.

Baselines are CMRDF-noVariants, Hierarchical-Cls, and CMRDF-KinOnly. They respectively probe the contributions of structured-variant training, distance semantics, and collision/ACM inputs.

Table 3: Zero-shot transfer under structured morphology shifts. Entries are macro-average percentages; Acc./Rec. denote SE(3) accuracy/Rec@ U_{nb} (higher is better).

Method	Limit-shift		Family-shift		Real-ODD	
	Acc.	Rec.	Acc.	Rec.	Acc.	Rec.
CMRDF-noVariants	84.7	76.5	79.6	68.7	42.8	21.4
Hierarchical-Cls	87.8	80.2	83.4	72.9	45.6	24.8
CMRDF-KinOnly	89.2	83.0	84.5	75.0	47.4	28.1
CMRDF	91.9	88.2	88.8	82.7	49.8	34.6

Table 3 supports three observations. First, CMRDF improves over CMRDF-noVariants across all three groups, with the largest gains on Family-shift, showing that structured-variant training improves morphology coverage. Second, adding collision geometry and ACM information consistently improves both SE3-Acc and $\text{Rec}@U_{\text{nb}}$, with the clearest gains on Family-shift and a stronger effect on recovery. Third, CMRDF outperforms Hierarchical-Cls across all three groups, particularly on $\text{Rec}@U_{\text{nb}}$, supporting the contribution of distance-based repair under morphology shifts. Real-ODD remains an out-of-support diagnostic with limited absolute performance; the zero-shot evidence therefore concerns joint-limit and same-family geometry shifts.

4.5 Downstream Data Utility for Robot Policy Transfer

Heterogeneous demonstrations may remain unreachable on the target morphology after coordinate alignment, limiting their direct use for supervised fine-tuning (SFT) of end-effector pose policies. We convert an in-house dataset of 845 single-arm demonstrations from A to B, two same-family seven-DoF serial manipulators with different link lengths and joint limits; B has a smaller reachable workspace. The demonstrations cover ten short-horizon supermarket tasks: shelf picking and tabletop flipping to turn the barcode upward for five SKUs. After common initial alignment, only 2 pass exact IK checks over the complete trajectory. We compare Align (alignment only), production NRT (tracking replay, holding the last valid command or using a nearby feasible pose when needed), Point-IK (a larger per-waypoint IK budget followed by the common servo to generate continuous trajectories), and CMRDF. CMRDF uses whole-trajectory reachability to adjust a shared task-frame transform, finding a feasible overall placement while preserving relative SE(3) motion and gripper timing. All methods share an execution stack and a fixed, method-independent set of 72 real-robot test trajectories; $\pi_{0.5}$ [27] SFT starts from the same 100 source demonstration IDs, with the same visual processing, model, sampling budget rule, and evaluation to compare native target demonstration requirements. This target-side data-utility study does not test zero-shot field transfer (Appendix B.6).

Table 4: Cross-morphology demonstration conversion and utility for policy learning.

Method	Outputs /845	Real-task success /72 $\uparrow$	Shape P95 (cm / $^\circ$) $\downarrow$	Tolerance violations (%) $\downarrow$	$K_{0.9\text{Ref}}$ $\downarrow$	One-time build (h)	Processing (s/trajectory) $\downarrow$
Align	2 (exact IK)	0	Reference	Reference	80	0	0.13
NRT	845 (commands)	27	7.7 / 6.4	59.3	65	0	4.05
Point-IK	469 (trajectories)	18	9.6 / 36.3	88.3	60	0	0.99
CMRDF	697 (validated trajectories)	59	1.5 / 2.9	8.5	20	6.5	0.38

Note: Shape P95 is the median of per-trajectory 95th-percentile residuals after removing one global rigid transform per trajectory, reported in position/orientation order. The tolerance violation rate is the fraction of waypoints with position residuals exceeding 3 cm or orientation residuals exceeding 10° . $K_{0.9\text{Ref}}$ is the smallest native demonstration count at which the mean real-task success rate over three seeds reaches 90% of that of a reference policy trained only on 100 native target demonstrations.

NRT produces commands for all demonstrations, but 59.3% of waypoints exceed the tolerance, showing that command output does not imply preservation of task-motion structure. Point-IK has an 88.3% tolerance violation rate after the common servo, showing that waypoint-wise feasibility does not ensure continuous-motion fidelity. CMRDF produces 697 independently validated complete trajectories, achieves 59/72 real-robot task successes, and has position/orientation shape P95 of 1.5 cm/2.9 $^\circ$. This advantage extends to policy learning: CMRDF needs only 20 native demonstrations to reach the same reference threshold, compared with 80, 65, and 60 for Align, NRT, and Point-IK, respectively, reducing native-data requirements to approximately 1/3–1/4 of those of the comparison methods.

5 Limitations

CMRDF models intrinsic pose reachability determined by joint limits and self-collision constraints. It does not include environmental obstacles, contact, dynamics, actuator constraints, or control trackability. It therefore provides a reachability prior rather than replacing scene-aware planning or safety-critical verification. The Position-SDF and Orientation-SDF are conditionally coupled proxies in meters and radians, not a single signed-distance field under a joint SE(3) metric. Their numerical targets depend on voxel resolution, sampling density, IK timeout, and warm-start quality. Reported full-pose IK budgets do not measure total construction or training cost. Zero-shot evidence is limited to joint-limit and same-family geometry shifts; broader cross-family, cross-DoF, and out-of-support transfer remains unestablished. Sequential projection is a local repair procedure: it does not guarantee a globally optimal task-weighted correction or replace global search, task-level replanning, or execution-time verification.

6 Conclusion and Future Work

We presented CMRDF, a morphology-conditioned hierarchical representation for cross-morphology pose reachability. CMRDF decomposes monolithic SE(3) reachability into Position-SDF and position-conditioned Orientation-SDF proxies, concentrating full-pose IK supervision near reachability-critical boundaries. Experiments show improved prediction over a capacity-matched monolithic baseline at matched full-pose IK budgets. Separate evaluations demonstrate low-distortion local repair, zero-shot transfer under joint-limit and same-family geometry shifts, and target-side data utility for policy fine-tuning. Future work will compose CMRDF with scene-conditioned reachability and generative robot policies for proposal filtering, reranking, and repair.

References

- [1] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, T. Jackson, S. Jesmonth, N. J. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, K. Lee, S. Levine, Y. Lu, U. Malla, D. Manjunath, I. Mordatch, O. Nachum, C. Parada, J. Peralta, E. Perez, K. Pertsch, J. Quiambao, K. Rao, M. S. Ryoo, G. Salazar, P. R. Sanketi, K. Sayed, J. Singh, S. Sontakke, A. Stone, C. Tan, H. T. Tran, V. Vanhoucke, S. Vega, Q. Vuong, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich. RT-1: Robotics transformer for real-world control at scale. In *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023. doi:10.15607/RSS.2023.XIX.025. URL <https://www.roboticsproceedings.org/rss19/p025.html>.
- [2] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, Q. Vuong, V. Vanhoucke, H. Tran, R. Soricut, A. Singh, J. Singh, P. Sermanet, P. R. Sanketi, G. Salazar, M. S. Ryoo, K. Reymann, K. Rao, K. Pertsch, I. Mordatch, H. Michalewski, Y. Lu, S. Levine, L. Lee, T.-W. E. Lee, I. Leal, Y. Kuang, D. Kalashnikov, R. Julian, N. J. Joshi, A. Irpan, B. Ichter, J. Hsu, A. Herzog, K. Hausman, K. Gopalakrishnan, C. Fu, P. Florence, C. Finn, K. A. Dubey, D. Driess, T. Ding, K. M. Choromanski, X. Chen, Y. Chebotar, J. Carbajal, N. Brown, A. Brohan, M. G. Arenas, and K. Han. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR, 2023. URL <https://proceedings.mlr.press/v229/zitkovich23a.html>.
- [3] Open X-Embodiment Collaboration. Open X-Embodiment: Robotic learning datasets and RT-X models. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6892–6903. IEEE, 2024. doi:10.1109/ICRA57147.2024.10611477. URL <https://ieeexplore.ieee.org/document/10611477>.
- [4] Octo Model Team, D. Ghosh, H. R. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, J. Luo, Y. L. Tan, L. Y. Chen, P. R. Sanketi, Q. Vuong, T. Xiao, D. Sadigh, C. Finn, and S. Levine. Octo: An open-source generalist robot policy. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024. doi:10.15607/RSS.2024.XX.090. URL <https://www.roboticsproceedings.org/rss20/p090.pdf>.
- [5] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn. OpenVLA: An open-source vision-language-action model. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 2679–2713. PMLR, 2025. URL <https://proceedings.mlr.press/v270/kim25c.html>.
- [6] F. Zacharias, C. Borst, and G. Hirzinger. Capturing robot workspace structure: Representing robot capabilities. In *2007 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3229–3236. IEEE, 2007. doi:10.1109/IROS.2007.4399105. URL <https://doi.org/10.1109/IROS.2007.4399105>.
- [7] F. Zacharias, C. Borst, S. Wolf, and G. Hirzinger. The capability map: A tool to analyze robot arm workspaces. *International Journal of Humanoid Robotics*, 10(4):1350031, 2013. doi:10.1142/S021984361350031X. URL <https://doi.org/10.1142/S021984361350031X>.
- [8] S. Kim and J. Perez. Learning reachable manifold and inverse mapping for a redundant robot manipulator. In *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pages 4731–4737. IEEE, 2021. doi:10.1109/ICRA48506.2021.9561589. URL <https://doi.org/10.1109/ICRA48506.2021.9561589>.
- [9] M. Murooka, I. Kumagai, M. Morisawa, and F. Kanehiro. Learning differentiable reachability maps for optimization-based humanoid motion generation. In *2025 IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids)*, pages 1–8. IEEE, 2025. doi:10.1109/

- Humanoids65713.2025.11203189. URL <https://doi.org/10.1109/Humanoids65713.2025.11203189>.
- [10] N. Vahrenkamp, T. Asfour, and R. Dillmann. Robot placement based on reachability inversion. In *2013 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1970–1975. IEEE, 2013. doi:10.1109/ICRA.2013.6630839. URL <https://doi.org/10.1109/ICRA.2013.6630839>.
 - [11] J. Dong and J. C. Trinkle. Orientation-based reachability map for robot base placement. In *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1488–1493. IEEE, 2015. doi:10.1109/IROS.2015.7353564. URL <https://doi.org/10.1109/IROS.2015.7353564>.
 - [12] A. Makhal and A. K. Goins. Reuleaux: Robot base placement by reachability analysis. In *2018 Second IEEE International Conference on Robotic Computing (IRC)*, pages 137–142. IEEE, 2018. doi:10.1109/IRC.2018.00028. URL <https://doi.org/10.1109/IRC.2018.00028>.
 - [13] M. Rudorfer. RM4D: A combined reachability and inverse reachability map for common 6-/7-Axis robot arms by dimensionality reduction to 4D. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 7689–7695. IEEE, 2025. doi:10.1109/ICRA55743.2025.11128095. URL <https://doi.org/10.1109/ICRA55743.2025.11128095>.
 - [14] L. Mescheder, M. Oechsle, M. Niemeyer, S. Nowozin, and A. Geiger. Occupancy networks: Learning 3D reconstruction in function space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4460–4470, 2019. doi:10.1109/CVPR.2019.00459. URL https://openaccess.thecvf.com/content_CVPR_2019/html/Mescheder_Occupancy_Networks_Learning_3D_Reconstruction_in_Function_Space_CVPR_2019_paper.html.
 - [15] J. J. Park, P. R. Florence, J. Straub, R. A. Newcombe, and S. Lovegrove. DeepSDF: Learning continuous signed distance functions for shape representation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 165–174, 2019. doi:10.1109/CVPR.2019.00025. URL https://openaccess.thecvf.com/content_CVPR_2019/html/Park_DeepSDF_Learning_Continuous_Signed_Distance_Functions_for_Shape_Representation_CVPR_2019_paper.html.
 - [16] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, and G. Wetzstein. Implicit neural representations with periodic activation functions. In *Advances in Neural Information Processing Systems*, volume 33, pages 7462–7473, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/53c04118df112c13a8c34b38343b9c10-Abstract.html>.
 - [17] A. Gropp, L. Yariv, N. Haim, M. Atzmon, and Y. Lipman. Implicit geometric regularization for learning shapes. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 3789–3799. PMLR, 2020. URL <https://proceedings.mlr.press/v119/gropp20a.html>.
 - [18] X. Chen, J. Külz, and M. Althoff. Generating robot capability maps with neural fields. In *RSS Workshop on Embodiment-Aware Robot Learning*, 2024. URL <https://openreview.net/forum?id=R0JGleLBRh>.
 - [19] C. Devin, A. Gupta, T. Darrell, P. Abbeel, and S. Levine. Learning modular neural network policies for multi-task and multi-robot transfer. In *2017 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2169–2176. IEEE, 2017. doi:10.1109/ICRA.2017.7989250. URL <https://doi.org/10.1109/ICRA.2017.7989250>.
 - [20] T. Wang, R. Liao, J. Ba, and S. Fidler. NerveNet: Learning structured policy with graph neural networks. In *International Conference on Learning Representations*, 2018. URL <https://openreview.net/forum?id=S1sqHMZCb>.

- [21] W. Huang, I. Mordatch, and D. Pathak. One policy to control them all: Shared modular policies for agent-agnostic control. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 4455–4464. PMLR, 2020. URL <https://proceedings.mlr.press/v119/huang20d.html>.
- [22] S. Hong, D. Yoon, and K.-E. Kim. Structure-aware transformer policy for inhomogeneous multi-task reinforcement learning. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=fy_XRVHqly.
- [23] Z. Xiong, J. Beck, and S. Whiteson. Universal morphology control via contextual modulation. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 38286–38300. PMLR, 2023. URL <https://proceedings.mlr.press/v202/xiong23a.html>.
- [24] G. Salhotra, I. A. Liu, and G. S. Sukhatme. Learning robot manipulation from cross-morphology demonstration. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2257–2277. PMLR, 2023. URL <https://proceedings.mlr.press/v229/salhotra23a.html>.
- [25] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. C. Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, pages 3942–3951. AAAI Press, 2018. doi:10.1609/aaai.v32i1.11671. URL <https://ojs.aaai.org/index.php/AAAI/article/view/11671>.
- [26] D. Coleman, I. A. Şucan, S. Chitta, and N. Correll. Reducing the barrier to entry of complex robotic software: a MoveIt! case study. *Journal of Software Engineering for Robotics*, 5(1): 3–16, May 2014. doi:10.6092/JOSER_2014_05_01_p3. URL <https://arxiv.org/abs/1404.3785>.
- [27] K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky. $\pi_{0.5}$: a vision-language-action model with open-world generalization. In *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 17–40. PMLR, 2025. URL <https://proceedings.mlr.press/v305/black25a.html>. Official code and model checkpoints: <https://github.com/Physical-Intelligence/openpi>.
- [28] C. R. Qi, H. Su, K. Mo, and L. J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 652–660. IEEE Computer Society, 2017. doi: 10.1109/CVPR.2017.16. URL https://openaccess.thecvf.com/content_cvpr_2017/html/Qi_PointNet_Deep_Learning_CVPR_2017_paper.html.
- [29] C. Ying, T. Cai, S. Luo, S. Zheng, G. Ke, D. He, Y. Shen, and T. Liu. Do transformers really perform badly for graph representation? In *Advances in Neural Information Processing Systems*, volume 34, pages 28877–28888, 2021. URL <https://proceedings.neurips.cc/paper/2021/hash/f1c1592588411002af340cbaedd6fc33-Abstract.html>.

A Methodological Details

A.1 Construction of Supervision Proxies

A.1.1 Position-SDF

Stage-I constructs a single-resolution voxel grid over the training-domain AABB of each morphology. Joint samples are generated by a progressive scrambled Sobol sequence and mapped to joint limits:

$$\xi_t \in [0, 1]^{d_{\mathbf{q}}}, \quad \mathbf{q}_t = \mathbf{q}_{\min} + \xi_t \odot (\mathbf{q}_{\max} - \mathbf{q}_{\min}). \quad (7)$$

Samples that pass self-collision checking are sent to FK, and hit voxels are marked as inside. To preserve the diversity of IK warm starts, each occupied voxel stores only a fixed-capacity representative reservoir, while `voxel_counts` records all FK hits. Voxel-level orientation coverage is used only as a statistic for anchor selection, local-context supervision, and training diagnostics; at inference time, the corresponding summary is not read from the held-out morphology, but predicted by the context head of the Position branch.

Once global FK coverage reaches a plateau, Stage-I switches to frontier completion. Let the current occupied voxel set be $\mathcal{V}_{\text{occ}}$. The frontier is defined as

$$\mathcal{F} = \{v \notin \mathcal{V}_{\text{occ}} | \mathcal{N}_6(v) \cap \mathcal{V}_{\text{occ}} \neq \emptyset\}. \quad (8)$$

For each $v \in \mathcal{F}$, warm-start joints are taken from the reservoirs of neighboring occupied voxels, and position-only DLS is performed on the voxel center and a small number of jittered targets:

$$\Delta \mathbf{q} = J_p^\top (J_p J_p^\top + \lambda_{\text{dls}}^2 I)^{-1} (p^* - p(\mathbf{q})). \quad (9)$$

After each update, joint-limit clamping and self-collision checking are applied. If the FK end-effector position falls into the target voxel, that voxel is marked as inside and its reservoir is updated. This stage verifies only frontier voxels immediately adjacent to known reachable regions, reducing cracks and thin-channel discontinuities caused by finite sampling, without turning all global outside voxels into an expensive search problem.

Afterward, only small-scale hole filling and closing are retained as post-processing. The method then traverses 6-neighborhood inside–outside voxel pairs and marks the inside side as the conservative boundary. Let $\mathcal{P}_\partial$ denote the set of boundary voxel centers. The Euclidean distance of a voxel center p_v is

$$d_\partial(v) = \min_{p_b \in \mathcal{P}_\partial} \|p_v - p_b\|_2. \quad (10)$$

The Position-SDF supervision proxy is defined as

$$\tilde{s}_p(p_v) = \begin{cases} 0, & \ell(v) = 0, \\ +d_\partial(v), & \ell(v) = 1, \\ -d_\partial(v), & \ell(v) = 2, \end{cases} \quad (11)$$

where $\ell = 0, 1, 2$ denote boundary, inside, and outside, respectively. This definition allows a small number of false negatives but explicitly suppresses false positive reachability.

Stage-I also constructs anchor-level supervision for the position-layer context head. For voxel centers, the local context target is formed from normalized Position-SDF values, finite-difference gradients, gradient magnitudes, local Laplacian or curvature proxies, FK hit density, and coverage summaries. If some statistics are not directly observable at final inference time, they are used only as supervision targets for the context head and are not used as external inputs to the Orientation-SDF decoder.

A.1.2 Orientation-SDF

Stage-II constructs orientation supervision only at selected position anchors. Anchors are selected from conservative inside and non-boundary voxels, and are sampled by importance according to

depth from the position boundary, coverage uncertainty, local coverage variation, and FK hit support. The geodesic angular distance for orientation consistently uses

$$d_{\text{SO}(3)}(u_1, u_2) = 2 \arccos(|\langle u_1, u_2 \rangle|) \in [0, \pi], \quad (12)$$

and short-arc interpolation is used in all slerp operations.

At each anchor p_a , CMRDF uses a Hopf-based intrinsic multiresolution grid over $\text{SO}(3)$. Each cell c has a center u_c , adjacency relations, a parent-child hierarchy, and a covering radius r_ℓ . Cell certification uses a local 7-point stencil:

$$\mathcal{S}(c) = \{u_c\} \cup \{\text{Exp}(\pm \rho_\ell e_j) u_c\}_{j=1}^3, \quad \rho_\ell = 0.5 r_\ell. \quad (13)$$

Each stencil query is labeled by a multi-start IK oracle as $y(u) \in \{+1, -1\}$. If the stencil is all positive, all negative, or contains mixed signs, the cell is marked as inside, outside, or mixed, respectively. Only mixed cells trigger local refinement, until the leaf-level radius or maximum depth is reached. The leaf graph is used only to discover adjacent certified inside/outside leaf-cell pairs, and is not used to define the final Orientation-SDF distance.

If all leaf cells are inside, the anchor becomes a full-reachable candidate. Progressive validation with early exit is then performed. If any probe fails, the failed probe and its nearest inside leaf center form an additional raw boundary bracket and are routed back into the regular-anchor workflow. Only when all validation probes pass is the anchor marked as full-reachable and written with full-positive supervision:

$$\tilde{\phi} = \tilde{\phi}_{\text{full}} = \pi, \quad \text{label} = +1. \quad (14)$$

This target represents positive lower-bound supervision, not an analytic distance to an empty boundary set.

For regular anchors, raw boundary brackets come from three sources: the nearest opposite-sign stencil pair inside a mixed leaf, adjacent certified inside/outside leaf-cell pairs, and failed-probe brackets from failed full validation. After geodesic clustering, representative refinement, and deduplication of raw brackets, we obtain the boundary representative set

$$\mathcal{B}_{\text{rep}}(p_a) = \{u_b^*\}. \quad (15)$$

Boundary samples are labeled with $\tilde{\phi} = 0$. Shell samples are generated from the reachable and infeasible ends of boundary representatives along short arcs. Global representative samples include certified inside/outside leaf centers and representative cell centers that pass homogeneous auditing. Their target distance is uniformly defined as

$$|\tilde{\phi}(u_g | p_a)| = \min_{u_b \in \mathcal{B}_{\text{rep}}(p_a)} d_{\text{SO}(3)}(u_g, u_b), \quad \tilde{\phi}(u_g | p_a) = y_g |\tilde{\phi}(u_g | p_a)|. \quad (16)$$

When the provisional global pool lacks sufficient hard negatives or hard positives, CMRDF performs outside-first homogeneous auditing, adding a small number of probes only to homogeneous cells that are sufficiently far from boundary representatives and useful for supplementing global supervision. If mixed signs appear during auditing, the cell is routed back into local refinement or the new-boundary-bracket workflow.

A.2 Query Encodings and Network Parameterization

Position queries are normalized by $L_{\text{ref}}(\mathcal{M})$, with $\tilde{p} = p/L_{\text{ref}}(\mathcal{M})$. The position encoding uses Fourier features:

$$E_p(\tilde{p}) = [\tilde{p}; \gamma(\tilde{p})], \quad \gamma(\tilde{p}) = [\sin(2^\ell \pi \tilde{p}), \cos(2^\ell \pi \tilde{p})]_{\ell=0}^{L-1}. \quad (17)$$

The orientation encoding uses a shared set of reference unit quaternions $\mathcal{U}_{\text{ref}} = \{u_k^{\text{ref}}\}_{k=1}^{K_R}$. For a unit quaternion u , define

$$\begin{aligned} a_k &= \text{clip}(|u^\top u_k^{\text{ref}}|, 0, 1), \\ c_k &= a_k, \\ s_k &= \sqrt{1 - a_k^2 + \epsilon}, \\ E_R(u) &= [c_1, s_1, \dots, c_{K_R}, s_{K_R}]. \end{aligned} \quad (18)$$

Here, a_k uses clipping to keep the cosine-like value within its numerical range. This encoding is insensitive to $u \sim -u$, and is index-aligned with $\text{SO}(3)$ sampling and the standardized orientation bank.

Morphology information is provided by the MorphologySpec encoder. MorphologySpec contains the kinematic tree, joint attributes, collision geometry, and ACM. A per-link PointNet [28] extracts collision geometry features, and a Graph Transformer [29] fuses kinematic topology, geometric relations, and allowed-collision relations to output the shared morphology embedding $m_{\mathcal{M}}$.

The Position branch uses a FiLM-SIREN backbone with a distance head and a local-context head [25, 16]:

$$h_p = f_{\theta_p}^{\text{trunk}}(E_p(\tilde{p}); m_{\mathcal{M}}), \quad \hat{s}_p = W_s h_p + b_s, \quad \hat{z}_{\text{pos}} = W_z h_p + b_z. \quad (19)$$

The physical distance is recovered by $\hat{s}_p = L_{\text{ref}}(\mathcal{M}) \hat{s}_p$. The predicted local position context $\hat{z}_{\text{pos}}$ is an anchor-level context prediction supervised by Stage-I position statistics; the center context and local patch context of the orientation layer are both sampled from this predicted field and its local neighborhood.

The Orientation decoder is a single-trunk FiLM-SIREN. Let $\tilde{s}_{\text{in}} = \text{clip}(\hat{s}_p, -1, 1)$, with center context $\hat{z}_c$ and local patch context $\hat{z}_l$. The base input is

$$\nu_0 = [E_R(u); E_p(\tilde{p}); \tilde{s}_{\text{in}}; \hat{z}_c; m_{\mathcal{M}}]. \quad (20)$$

FiLM parameters are generated from the local context and morphology embedding:

$$(\gamma_l, \beta_l)_{l=0}^{L_R-1} = G_{\text{film}}([\hat{z}_l; \hat{z}_c; m_{\mathcal{M}}]). \quad (21)$$

The trunk outputs

$$\hat{\phi}_{\text{raw}} = f_{\theta_R}(\nu_0; \{\gamma_l, \beta_l\}_{l=0}^{L_R-1}), \quad \hat{\phi} = \pi \tanh\left(\frac{\hat{\phi}_{\text{raw}}}{\pi}\right). \quad (22)$$

The coverage auxiliary head predicts the anchor-level orientation support ratio $\hat{\rho}_a$ and is used only for training losses and diagnostics; the logging-only sign probe does not enter the total loss, checkpoint selection, reachability decision, or pose repair.

A.3 Losses and Stratified Sampling

A.3.1 Position-SDF

For the Position-SDF proxy target, TSDF truncation and scale normalization are first applied:

$$y = \text{clip}(\tilde{s}_p, -s_{\text{trunc}}, s_{\text{trunc}}), \quad \bar{y} = \frac{y}{L_{\text{ref}}(\mathcal{M})}. \quad (23)$$

The position-layer loss is

$$\mathcal{L}_p = \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{sign}} \mathcal{L}_{\text{sign}} + \lambda_{\partial} \mathcal{L}_{\partial} + \lambda_{\text{ctx}} \mathcal{L}_{\text{ctx}} + \lambda_{\text{eik}} \mathcal{L}_{\text{eik}}. \quad (24)$$

$\mathcal{L}_{\text{reg}}$ regresses the truncated signed distance, $\mathcal{L}_{\text{sign}}$ uses an outside-heavy conservative sign loss, $\mathcal{L}_{\partial}$ anchors boundary voxels to the zero level set, $\mathcal{L}_{\text{ctx}}$ supervises the local-context head, and $\mathcal{L}_{\text{eik}}$ uses near-boundary equality together with a far-field one-sided Lipschitz penalty. Training sampling is stratified into boundary and non-boundary samples; non-boundary samples are further sampled by $\log|s|$ bucket, spatial bucket, and sign balance.

A.3.2 Orientation-SDF

The orientation-layer loss is grouped by function as follows:

$$\mathcal{L}_R = \mathcal{L}_R^{\text{sdf}} + \lambda_{\text{sign}} \mathcal{L}_R^{\text{sign}} + \lambda_{\text{one}} \mathcal{L}_R^{\text{one}} + \lambda_{\text{ctx}} \mathcal{L}_R^{\text{ctx}} + \lambda_{\text{eik}} \mathcal{L}_R^{\text{eik}}. \quad (25)$$

$\mathcal{L}_R^{\text{sdf}}$ includes Huber regression on mined samples and the standardized orientation bank, where mined samples cover boundary, shell, and global representatives; boundary-zero supervision is included in

this group to stabilize the zero level set. $\mathcal{L}_R^{\text{sign}}$ directly acts on the sign logit of the main raw field, and uses balanced sign supervision on shell, global, and standardized masked queries. $\mathcal{L}_R^{\text{one}}$ covers the positive lower-bound term for full-reachable anchors, the far-negative conservative term, and hard inside/outside margins. $\mathcal{L}_R^{\text{ctx}}$ constrains the expression of orientation support range and cross-anchor variation by the local context, including coverage auxiliary, shell-pair separation and centering, and neighbor consistency. $\mathcal{L}_R^{\text{eik}}$ imposes near-boundary unit-gradient and far-field one-sided Lipschitz regularization on the raw field in local Lie-algebra coordinates of $\text{SO}(3)$.

For full-reachable anchors, ordinary boundary regression is not applicable. Training only requires $\hat{\phi}$ to remain sufficiently positive, and constrains the high-coverage state through the coverage auxiliary. For regular anchors, shell-pair and neighbor consistency are used only as training regularizers and do not change the final query interface.

A.4 Concrete Updates for Pose Repair

The local pose-repair experiments in Secs. 4.3–4.4 use sequential projection. The position update is performed in physical coordinates using the denormalized Position-SDF:

$$\hat{s}_p(p \mid \mathcal{M}) = L_{\text{ref}}(\mathcal{M}) \hat{s}_p(\tilde{p} \mid \mathcal{M}). \quad (26)$$

If the current position has not yet satisfied the reachable inner band $\hat{s}_p(p_t \mid \mathcal{M}) \geq \tau_p$, the orientation R_t is fixed and the position is updated along the position-field gradient:

$$p_{t+1} = p_t + \eta_p \frac{\tau_p - \hat{s}_p(p_t \mid \mathcal{M})}{\|\nabla_p \hat{s}_p(p_t \mid \mathcal{M})\|_2 + \epsilon} \nabla_p \hat{s}_p(p_t \mid \mathcal{M}). \quad (27)$$

Once the position satisfies the inner band but the orientation remains infeasible, orientation is updated in the Lie algebra. Let

$$g_{\omega,t} = \nabla_{\Delta\omega} \hat{\phi}_{\text{raw}}(\text{Exp}([\Delta\omega]_{\times}) R_t \mid p_t; \mathcal{M}) \Big|_{\Delta\omega=0}, \quad (28)$$

then

$$\Delta\omega_t = \eta_R \frac{-\hat{\phi}_{\text{raw}}(R_t \mid p_t; \mathcal{M})}{\|g_{\omega,t}\|_2 + \epsilon} g_{\omega,t}, \quad R_{t+1} = \text{Exp}([\Delta\omega_t]_{\times}) R_t. \quad (29)$$

This sequential projection is the online local pose-repair procedure evaluated in Secs. 4.3–4.4. Task weights are used only for the evaluation metric and the high-budget reference solution in Sec. 4.3, and do not change the update rules above. Sequential projection is the default because it is consistent with the position-first reachability semantics of CMRDF and avoids requiring the Orientation-SDF to interpret invalid orientation constraints when the position is infeasible.

B Supplementary Experimental Results

B.1 Baseline Reimplementation and Evaluation Splits

Table 1(a) evaluates reimplemented differentiable reachability maps (DRM) [9] with neural-network and radial-basis-function SVM learners, together with the CMRDF anchor-budget sweep. Table 1(b) reports the capacity-matched DeepMono comparison. For each DRM learner, changing MonoData to HierData evaluates supervision allocation; its comparison with CMRDF does not isolate architecture or model capacity.

The CMRDF anchor-budget settings use 2,048, 1,640, and 1,024 orientation anchors, corresponding to full-pose IK budgets of $1.00\times$, $0.79\times$, and $0.51\times$. Position supervision, architecture, training objectives, and test sets are unchanged. Anchor-Seen uses held-out orientations at supervised anchors; Anchor-Heldout uses positions excluded from orientation supervision. Both evaluate training morphologies with the common decision oracle. The DRM and anchor-budget summaries in Table 1(a) and Table 5 are reported separately from the three-seed capacity-matched summary in Table 1(b).

Table 5: Supplementary per-base Anchor-Heldout accuracy (%) for DRM and CMRDF.

Base morphology	DRM baselines				CMRDF variants		
	NN Mono	NN Hier	RSVM Mono	RSVM Hier	CMRDF 1.00×	CMRDF 0.79×	CMRDF 0.51×
AUBO-i5	63.42	65.62	63.91	66.24	86.54	85.16	69.18
RealMan RM65	61.18	64.21	61.72	64.85	85.85	82.94	67.82
Franka Research 3	64.25	66.50	64.36	67.11	89.71	85.65	71.03
Kinova Gen3	60.74	63.64	61.04	64.72	84.47	82.77	66.59
Universal Robots UR5e	63.26	67.18	64.82	67.23	89.33	85.08	72.73
Macro avg.	62.57	65.43	63.17	66.03	87.18	84.32	69.47

B.2 Per-Base Results for the DRM and Anchor-Budget Evaluation

Table 5 provides the per-base NB-Heldout results underlying the DRM and anchor-budget evaluation in Table 1(a) and Figure 4(a). CMRDF exceeds the evaluated DRM baselines on each base at all three listed budgets. These per-base results do not establish a per-base advantage over DeepMono; the capacity-matched evidence in Table 1(b) is reported as three-seed summaries of macro averages.

B.3 Construction and Evaluation Protocol of the Pose Repair Benchmark

The repair benchmark in Sec. 4.3 constructs near-boundary failures from held-out feasible seed poses, rather than sampling infeasible poses globally at random. Let the seed pose be $x^+ = (p^+, R^+)$. We retain only seeds that are close to the position boundary and have a finite orientation boundary, and generate two types of queries from them. The first type is **Ori-only**: the position is kept unchanged, and the orientation is replaced by a negative shell orientation at the same anchor,

$$x_{\text{ori}} = (p^+, R^-), \quad |\tilde{\phi}(R^- | p^+)| \in \{0.04, 0.08\} \text{ rad.} \quad (30)$$

The second type is **Mixed**: the seed is first pushed outside the reachable position domain along the outward normal of the position field, and the same orientation perturbation is then applied,

$$n_{\text{out}}(p^+) = -\frac{\nabla \tilde{s}_p(p^+)}{\|\nabla \tilde{s}_p(p^+)\|_2 + \epsilon}, \quad p^- = p^+ + \epsilon_{\text{out}} n_{\text{out}}(p^+), \quad \epsilon_{\text{out}} \in \{h, 2h\}, \quad (31)$$

$$x_{\text{mix}} = (p^-, R^-). \quad (32)$$

For each base morphology, we sample fixed sets of 2000 **Ori-only** queries and 2000 **Mixed** queries, and recheck their oracle-infeasible labels with the same oracle as in the main text.

Evaluation uses two canonical task regimes:

$$w_p : w_R = 4 : 1, \quad w_p : w_R = 1 : 4. \quad (33)$$

These weights are used to define the evaluation metric and the high-budget reference solution, and are not input to the CMRDF sequential projection update rules. The unified repair cost is defined as

$$J_{\text{norm}}(\bar{x}; x_0) = \frac{w_p}{2} \left\| \frac{\bar{p} - p_0}{L_{\text{ref}}} \right\|_2^2 + \frac{w_R}{2} d_{\text{SO}(3)}(\bar{R}, R_0)^2, \quad (34)$$

and

$$\text{Gap-to-Ref} = J_{\text{norm}}(\bar{x}_{\text{method}}; x_0) - J_{\text{norm}}(\bar{x}_{\text{ref}}; x_0). \quad (35)$$

Online methods include **DRM-Score**, **Hierarchical-Cls**, **Search-IK**, **CMRDF w/o Ori-Eik**, and **CMRDF**; **Ref-IK** is used only as a high-budget reference solution. The final feasibility of all methods is verified by the same evaluation oracle, and latency does not include final validation time.

Table 6: Additional breakdown of the pose repair benchmark. Ori-only evaluates unnecessary position drift during orientation repair; Mixed evaluates low-distortion recovery when translation and rotation are both perturbed. Values are averaged over two task regimes.

Method	Ori-only		Mixed			Time
	Succ. (%) $\uparrow$	$\Delta p/L_{\text{ref}} \downarrow$	Succ. (%) $\uparrow$	$J_{\text{norm}} \downarrow$	Gap $\downarrow$	ms $\downarrow$
DRM-Score	86.6	0.049	84.1	0.0376	0.0259	1.52
Hierarchical-Cls	89.2	0.038	87.4	0.0313	0.0196	1.33
Search-IK	94.6	0.019	91.5	0.0187	0.0070	13.84
CMRDF w/o Ori-Eik	93.7	0.013	90.7	0.0152	0.0035	1.43
CMRDF	95.8	0.007	93.5	0.0125	0.0008	1.31
Ref-IK (high-budget)	98.9	0.005	95.2	0.0117	0.0000	34.57

B.4 Additional Breakdown of the Pose Repair Benchmark by Query Family

The breakdown results are consistent with the main-text conclusions. For Ori-only, the advantage of CMRDF mainly lies in smaller unnecessary position drift. For Mixed, it performs best among the evaluated online methods in success rate, repair cost, and Gap-to-Ref, indicating that it is not merely easier for CMRDF to find feasible solutions, but that it also keeps repair distortion low under the task-weighted evaluation metric.

B.5 Protocol for Structured-Shift Zero-Shot Transfer

Sec. 4.4 uses a target-balanced protocol. The three target groups are as follows: Limit-shift, consisting of 2 held-out joint-limit variants from each of the 5 base morphologies, for a total of 10 targets; Family-shift, consisting of 2 held-out same-family geometry variants from each of the 5 base morphologies, for a total of 10 targets; and Real-OOD, consisting of 2 real held-out morphologies. Except for CMRDF-noVariants, all shared models are trained on the 35 training morphologies in Sec. 4.1, but do not access these targets. CMRDF-noVariants is trained only on the 5 real base morphologies and is used to test the contribution of structured variants.

For each target, we independently construct a test set containing equal numbers of four query types: reachable interior, reachable near-boundary, unreachable near-boundary, and unreachable farther-out. The overall prediction metric is defined as

$$\text{SE3-Acc}(\mathcal{M}_t) = \frac{1}{|Q_{\text{cls}}(\mathcal{M}_t)|} \sum_{x \in Q_{\text{cls}}(\mathcal{M}_t)} \mathbf{1}[\hat{y}(x) = y_{\text{oracle}}(x)]. \quad (36)$$

For the oracle-infeasible near-boundary subset $Q_{U,nb}(\mathcal{M}_t)$, we further execute the same sequential projection as in Sec. 4.3 and report the end-to-end recovery rate

$$\text{Rec@}U_{nb}(\mathcal{M}_t) = \frac{1}{|Q_{U,nb}(\mathcal{M}_t)|} \sum_{x \in Q_{U,nb}(\mathcal{M}_t)} \mathbf{1}[\hat{y}(x) = 0] \mathbf{1}[\text{oracle}(\bar{x}) = 1]. \quad (37)$$

This metric simultaneously requires first correctly rejecting the query and then repairing it back to a reachable pose.

All four methods share the same target pools and evaluation oracle. This evaluation focuses on whether the method makes correct decisions on the target morphology and whether a query predicted as infeasible can be repaired back to reachability. It does not report task-weighted repair cost.

B.6 Downstream Data Utility: Protocol and Additional Results

B.6.1 Morphologies, Tasks, and Demonstration Conversion

Morphologies and data. Source morphology A and target morphology B are seven-DoF serial manipulators from the same family, differing in link lengths and joint limits. B has shorter links

overall and a smaller reachable workspace, and the target robot is equipped with a single-DoF gripper. The experiment converts demonstrations from A to B and evaluates their data utility for trajectory replay and policy learning on B.

Our in-house dataset contains 845 single-arm demonstrations. The five SKUs cover three packaging types—tubs, bags, and boxes—with different visual features and corresponding grasp poses and flipping methods. Each SKU has two separate operations, shelf picking and tabletop flipping, with the latter aiming to turn the barcode upward. Together, these form ten short-horizon single-arm tasks.

Common input. All source trajectories first undergo the same base-frame, height, and orientation alignment, providing a common input to the four processing pipelines. Demonstration conversion, real-robot execution, and policy learning evaluate data outputs, task completion, and native-demonstration requirements, respectively. The fixed real-robot execution test uses 72 trajectories, while policy learning starts from the same 100 source demonstration IDs shared across methods.

Align. This pipeline retains only the trajectories after the common initial alignment, without further adjustment for target reachability. The 2/845 entry in Table 4 is the number of demonstrations that pass exact IK checks over the complete trajectory after this processing.

Production NRT. This pipeline uses tracking-based replay on the target robot. When no valid solution is available for a target pose, it holds the last valid command or uses a nearby feasible pose. Its outputs are therefore command sequences that may include hold commands.

Point-IK. This pipeline allocates a larger IK budget to each waypoint, then passes the discrete results to the common production servo to generate a continuous trajectory. It handles waypoint feasibility individually and does not optimize a task-frame placement shared by the entire trajectory.

CMRDF. This pipeline evaluates the entire trajectory using the reachability field for the target morphology and adjusts a task-frame transform shared by all waypoints. Let the demonstration after initial alignment be

$$\mathcal{T} = \{(T_t, g_t, \tau_t)\}_{t=1}^L, \quad (38)$$

where $T_t \in SE(3)$ is the end-effector pose, g_t is the gripper command, τ_t is the corresponding time, and L is the trajectory length. Let $G \in SE(3)$ denote the shared task-frame adjustment. The transformed reference trajectory satisfies

$$\tilde{T}_t = GT_t, \quad \tilde{g}_t = g_t, \quad \tilde{\tau}_t = \tau_t. \quad (39)$$

For any two waypoints t and s , we have

$$\tilde{T}_t^{-1}\tilde{T}_s = (GT_t)^{-1}(GT_s) = T_t^{-1}T_s. \quad (40)$$

Thus, the shared rigid adjustment preserves the relative poses within the reference trajectory and the correspondence between gripper commands and action times. CMRDF adjusts the task-frame placement of the entire demonstration rather than applying the local pose repair in Appendix A.4 independently to each waypoint. Converted complete trajectories are counted as valid outputs after independent validation; motion fidelity after execution is evaluated using the metrics in Appendix B.6.2.

B.6.2 Output Statistics, Real-Robot Evaluation, and Motion Fidelity

Output statistics. Table 4 reports conversion outputs and real-task success separately. NRT’s 845 outputs are command sequences, Point-IK’s 469 outputs are trajectories, and CMRDF’s 697 outputs are independently validated complete trajectories. These counts refer to different output types and are therefore not directly compared as valid trajectory rates under a common definition.

Task success criteria. Success in each trial is judged according to the specified SKU and task type. Shelf picking requires the robot to remove the target item from the shelf using the grasp pose for

that SKU; tabletop flipping requires it to use the grasping and flipping method for that SKU to turn the target item’s barcode upward. The two operations are evaluated separately as independent tasks. Success is determined by these operational requirements and completion states, not by IK success or compliance with trajectory shape tolerances.

Real-robot evaluation of the converted trajectories uses the same target-side execution stack and a method-independent set of 72 test trajectories. All methods report task success counts with 72 as the denominator. These results evaluate actual execution of the converted trajectories and are reported separately from the policy fine-tuning evaluation in Appendix B.6.3.

Trajectory shape error. Shape evaluation removes one global rigid transform per trajectory to distinguish changes in overall placement from distortion within the trajectory. Let the waypoints of reference trajectory j be $(p_{j,t}, R_{j,t})$, and let the trajectory being evaluated after global rigid alignment be $(\bar{p}_{j,t}, \bar{R}_{j,t})$. We define the position and orientation residuals as

$$e_{j,t}^p = \|\bar{p}_{j,t} - p_{j,t}\|_2, \quad e_{j,t}^R = d_{SO(3)}(\bar{R}_{j,t}, R_{j,t}). \quad (41)$$

We compute the 95th percentile of waypoint residuals within each trajectory, then take the median over the set of evaluated demonstrations $\mathcal{E}$:

$$\begin{aligned} \text{ShapeP95}_p &= \text{median}_{j \in \mathcal{E}} [\text{P95}_t(e_{j,t}^p)], \\ \text{ShapeP95}_R &= \text{median}_{j \in \mathcal{E}} [\text{P95}_t(e_{j,t}^R)]. \end{aligned} \quad (42)$$

Position and orientation errors are reported in centimeters and degrees, respectively. The tolerance violation rate is the fraction of waypoints whose position residual exceeds 3 cm or whose orientation residual exceeds 10° ; either condition counts as a violation. Align serves as the shape reference. Global rigid alignment during evaluation is used only to compute residuals and does not alter the trajectory sent to the robot. It is also distinct from the shared task-frame adjustment used for trajectory conversion in Appendix B.6.1.

Stage-wise breakdown of Point-IK. Table 7 compares the shape errors of the discrete IK results and the output of the common servo controller.

Table 7: Trajectory shape errors of Point-IK at the discrete IK and servo stages.

Evaluation stage	Position shape P95 (cm) ↓	Orientation shape P95 ($^\circ$) ↓	Tolerance violation rate (%) ↓
Discrete IK results	5.9	1.0	41.1
After the common servo controller	9.6	36.3	88.3

Note: Shape P95 and the tolerance violation rate use the definitions in Table 4; the Point-IK row in Table 4 reports the servo-stage results.

The shape fidelity obtained by discrete pose solving is not maintained during continuous execution. This difference shows that, under the execution pipeline used here, improving waypoint-wise feasibility alone does not ensure continuous-motion fidelity. We therefore report conversion outputs, post-execution trajectory shape, and real-task success as complementary measures.

B.6.3 VLA Fine-Tuning and Native Demonstration Budgets

Training data and common settings. All methods use the same 100 source demonstration IDs from morphology A as conversion inputs. The processed source demonstrations are mixed with native demonstrations from B for $\pi_{0.5}$ [27] SFT, and samples whose conversion fails are excluded from training. All methods use the same visual preprocessing, policy model, optimization settings, and rule for the data exposure budget. The reference policy is trained only on 100 native demonstrations from B.

Observation and action representations. The model receives two RGB views, the robot state, and a task instruction at the current time, without stacking past observations. All images are resized with their aspect ratio preserved and padded to 224×224 . The single-arm robot state is written as

$$z_t = [p_t, \rho(R_t), g_t] \in \mathbb{R}^{10}, \quad (43)$$

where $p_t \in \mathbb{R}^3$ is the end-effector position, $\rho(R_t) \in \mathbb{R}^6$ is the Rot6D encoding formed from the first two columns of the rotation matrix, and g_t is the continuous gripper state. The action chunk uses the current observed state as a shared anchor. Before normalization, the supervision target at step h is

$$a_{t,h} = [p_{t+h} - p_t, \rho(R_{t+h}) - \rho(R_t), g_{t+h}], \quad h = 1, \dots, 30. \quad (44)$$

The position and orientation encodings use differences from the same anchor, while the gripper state remains absolute. Each sample starts from one recorded observation frame and supervises the next 30 action steps; any missing action steps at the end of a trajectory are padded with that trajectory’s final state. States and actions use quantile-based normalization, and dimensions padded to match the model interface do not contribute to the action loss.

Optimization and data exposure. All methods use the common optimization settings in Table 8, without a completion head. Let $N_{m,k}$ be the number of recorded observation frames in the actual training set for method m at native-demonstration budget k , and let $M_{m,k}$ be the cumulative number of sampled training chunks. The training budget follows the same rule for all methods:

$$\frac{M_{m,k}}{N_{m,k}} = 15 \quad (45)$$

This budget measures total exposure through sampling and does not require every recorded frame to be sampled exactly 15 times; final-state padding does not increase $N_{m,k}$. The effective global batch size is 4 chunks.

Table 8: Common optimization settings and training budget for VLA fine-tuning.

Item	Setting
Main-model optimizer	AdamW
β_1, β_2	0.9, 0.95
ϵ	10^{-8}
Weight decay	10^{-10}
Global gradient clipping threshold	1.0
Main-model learning rate	5×10^{-5} , subsequently reduced to 1×10^{-5}
Warmup	200 optimizer update steps
Effective global batch size	4 chunks
Action horizon	30 steps
Total data exposure budget	15 times the number of recorded observation frames in the actual training set
Completion head	Not used

Action decoding and execution. At inference time, the outputs are first denormalized, then each future target is recovered using the state from the same observation, without cumulative summation over the predicted chunk. The recovered Rot6D encoding is orthogonalized to obtain a rotation matrix and converted to the orientation representation required by the execution interface. All methods use the same SDK version and the same procedure for sending end-effector pose commands.

Native demonstration budget metric. The fine-tuned policies are executed on morphology B, with the task success criteria in Appendix B.6.2. Let $\mathcal{K}$ be the set of evaluated native-demonstration budgets, $S_{m,s}(k)$ the real-task success rate of method m at budget $k \in \mathcal{K}$ with seed s , and S_{Ref}

the success rate of the reference policy trained only on 100 native target demonstrations. We first compute the mean success rate over three seeds,

$$\bar{S}_m(k) = \frac{1}{3} \sum_{s=1}^3 S_{m,s}(k), \quad (46)$$

then define the smallest native-demonstration budget required to reach the reference performance threshold as

$$K_{0.9\text{Ref}}^{(m)} = \min \{k \in \mathcal{K} : \bar{S}_m(k) \geq 0.9S_{\text{Ref}}\}. \quad (47)$$

This threshold is 90% of the reference policy’s success rate, not an absolute task success rate of 90%. The $K_{0.9\text{Ref}}$ values for Align, NRT, Point-IK, and CMRDF in Table 4 are 80, 65, 60, and 20, respectively.

B.6.4 One-Time Construction and Trajectory Processing Costs

Table 4 reports the one-time construction of the reachability field separately from the processing time per trajectory. CMRDF construction takes 6.5 h in total, including supervision generation and reachability model training; training takes approximately 3 h on four identical accelerators. After construction, the marginal processing time is 0.38 s per trajectory, compared with 0.13 s, 4.05 s, and 0.99 s for Align, NRT, and Point-IK, respectively.

The constructed reachability field can be reused for subsequent trajectory processing. CMRDF’s cost therefore consists of one-time construction and subsequent per-trajectory processing. This experiment reports a lower marginal processing cost, not a reduction in total compute for the corpus of 845 demonstrations.